

CODE:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.util.ArrayList;
```

```
// 🏦 Account Class
```

```
class Account {
    String accNo, pin;
    double balance;
    ArrayList<String> history = new ArrayList<>();

    Account(String accNo, String pin, double balance) {
        this.accNo = accNo;
        this.pin = pin;
        this.balance = balance;
    }
}
```

```
public class ATMSysGUI extends JFrame {
```

```
    CardLayout card = new CardLayout();
    JPanel mainPanel = new JPanel(card);
```

```
    ArrayList<Account> accounts = new ArrayList<>();
    Account currentUser;
```

```
// 🗝️ Login Fields
```

```
    JTextField accField;
    JPasswordField pinField;
```

```

// 💰 Dashboard Components

JLabel balanceLabel;

JTextArea statementArea;

public ATMSystemGUI() {
    setTitle("ATM System");
    setSize(800, 500);
    setDefaultCloseOperation(EXIT_ON_CLOSE);
    setLocationRelativeTo(null);

    // Sample Account
    accounts.add(new Account("1234", "1111", 5000));

    mainPanel.add(loginPanel(), "login");
    mainPanel.add(dashboardPanel(), "dashboard");

    add(mainPanel);
    card.show(mainPanel, "login");

    setVisible(true);
}

// 🗝️ Login Panel
JPanel loginPanel() {
    JPanel panel = new JPanel(new GridBagLayout());
    panel.setBackground(new Color(30, 30, 45));

    GridBagConstraints g = new GridBagConstraints();
    g.insets = new Insets(10, 10, 10, 10);

    accField = new JTextField(15);

```

```

pinField = new JPasswordField(15);

JButton loginBtn = new JButton("Login");

g.gridx = 0; g.gridy = 0;
panel.add(new JLabel("Account No:"), g);
g.gridx = 1;
panel.add(accField, g);

g.gridx = 0; g.gridy = 1;
panel.add(new JLabel("PIN:"), g);
g.gridx = 1;
panel.add(pinField, g);

g.gridy = 2; g.gridx = 1;
panel.add(loginBtn, g);

loginBtn.addActionListener(e -> login());

return panel;
}

// 🏠 Dashboard Panel (Web-like UI)
JPanel dashboardPanel() {
    JPanel panel = new JPanel(new BorderLayout());
    panel.setBackground(new Color(40, 40, 60));

    // Sidebar
    JPanel side = new JPanel(new GridLayout(6,1,10,10));
    side.setBackground(new Color(50,50,70));

```

```

JButton balBtn = new JButton("Balance");
JButton depBtn = new JButton("Deposit");
JButton witBtn = new JButton("Withdraw");
JButton stmtBtn = new JButton("Mini Statement");
JButton logoutBtn = new JButton("Logout");

JButton[] btns = {balBtn, depBtn, witBtn, stmtBtn, logoutBtn};
for(JButton b: btns){
    b.setFocusPainted(false);
    side.add(b);
}

// Main Area
JPanel content = new JPanel(new CardLayout());
content.setBackground(new Color(30,30,45));

// Balance Panel
JPanel balPanel = new JPanel();
balanceLabel = new JLabel("Balance: ₹0");
balanceLabel.setFont(new Font("Arial", Font.BOLD, 20));
balPanel.add(balanceLabel);

// Statement Panel
JPanel stmtPanel = new JPanel(new BorderLayout());
statementArea = new JTextArea();
stmtPanel.add(new JScrollPane(statementArea));

content.add(balPanel, "balance");
content.add(stmtPanel, "statement");

CardLayout contentCard = (CardLayout) content.getLayout();
```

```

// Button Actions

balBtn.addActionListener(e -> {

    balanceLabel.setText("Balance: ₹" + currentUser.balance);

    contentCard.show(content, "balance");

});

depBtn.addActionListener(e -> deposit());

witBtn.addActionListener(e -> withdraw());

stmtBtn.addActionListener(e -> {

    statementArea.setText("");

    for(String s : currentUser.history){

        statementArea.append(s + "\n");

    }

    contentCard.show(content, "statement");

});

logoutBtn.addActionListener(e -> card.show(mainPanel, "login"));

panel.add(side, BorderLayout.WEST);

panel.add(content, BorderLayout.CENTER);

return panel;
}

// 🗝 Login Logic

void login() {

    String acc = accField.getText();

    String pin = new String(pinField.getPassword());

```

```

for(Account a : accounts){
    if(a.accNo.equals(acc) && a.pin.equals(pin)){
        currentUser = a;
        card.show(mainPanel, "dashboard");
        return;
    }
}

JOptionPane.showMessageDialog(this, "Invalid Login!");
}

```

// 💰 Deposit

```

void deposit() {
    String amtStr = JOptionPane.showInputDialog("Enter amount:");
    if(amtStr == null) return;

    double amt = Double.parseDouble(amtStr);
    if(amt <= 0){
        JOptionPane.showMessageDialog(this, "Invalid amount");
        return;
    }

    currentUser.balance += amt;
    currentUser.history.add("Deposited ₹" + amt);
    JOptionPane.showMessageDialog(this, "Deposit Successful!");
}

```

// 💸 Withdraw

```

void withdraw() {
    String amtStr = JOptionPane.showInputDialog("Enter amount:");
    if(amtStr == null) return;

```

```
double amt = Double.parseDouble(amtStr);

if(amt > currentUser.balance){
    JOptionPane.showMessageDialog(this,"Insufficient Balance!");
    return;
}

currentUser.balance -= amt;
currentUser.history.add("Withdrawn ₹" + amt);
JOptionPane.showMessageDialog(this,"Withdraw Successful!");
}

public static void main(String[] args) {
    new ATMSystemGUI();
}
}
```